Data Mining
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[13: Dimensionality Reduction (2)]

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Motivation for Autoencoder

- autoencoders matter because they transform high-dimensional data into a compact latent representation through the bottleneck
  - latent code is not only compressed, but also informative and useful for data analysis
  - it preserves the most salient structure of the input while discarding irrelevant variation
  - information-theoretic view: autoencoders seek a representation that is as compact as possible while still retaining essential information for reconstruction
  - data mining view: autoencoders help uncover low-dimensional structure hidden in complex high-dimensional data
  - robust learning: they reduce redundancy and suppress noise, leading to cleaner and more stable representations
  - practical utility: the learned latent features can support downstream tasks such as clustering, visualization, classification, and anomaly detection

Autoencoder as a Compression System

- an autoencoder consists of three components:

$$X \xrightarrow{f_\theta} Z \xrightarrow{g_\phi} \hat{X}$$

  - the encoder maps the input X to a latent representation Z
  - the decoder reconstructs an approximation $\hat{X}$ from Z
  - the bottleneck constraints the latent code and limits information flow

- why is the bottleneck important?
  - it forces the model to represent the input in a more compact form
  - only information that passes through Z can be used for reconstruction
  - as a result, the network is encouraged to preserve salient structure rather than memorize every detail

- in this sense, an autocndoer is a form of lossy compression:
  - Z acts as a compressed code of the input
  - reconstruction quality depends on how much useful information is retained in Z
  - the learning problem is to compress X while still reconstructing it well

Recall: Entropy

- entropy as uncertainty:

$$H(X) = -\sum_{x \in \mathcal{X}} p(x) \log p(x)$$

  - entropy quantifies how uncertain or unpredictable X is
  - higher entropy generally implies more information is needed to describe the source

- conditional entropy as remaining uncertainty

$$H(X|Z) = -\sum_{x,z} p(x,z) \log p(x|z)$$

  - this measures how uncertain X remains after observing Z
  - if Z is highly informative about X, then $H(X|Z)$ is small

- connection to representation learning
  - in an AE, the latent code Z should capture information that helps reconstruct X
  - a useful representation reduces uncertainty about the input while using limited capacity

Kullback-Leibler Divergence and Mutual Information

- KL divergence measures how different one distribution is from another:

$$D_{KL}(P||Q) = \sum_x p(x) \log \frac{p(x)}{q(x)} \geq 0$$

  - asymmetric: $D_{KL}(P||Q) \neq D_{KL}(Q||P)$
  - equal zero if and only if $P = Q$

- mutual information measures how much information two random variables share:

$$I(X;Z) = H(X) - H(X|Z) = H(Z) - H(Z|X)$$

$$= D_{KL}(p(x,z)||p(x)p(z)) \geq 0$$

  - if X and Z are independent, then $I(X;Z) = 0$
  - the larger $I(X;Z)$ is, the more informative Z is about X
  - equivalently, mutual information measures how much knowing Z reduces uncertainty about X

- why does this matter for AE?
  - the latent code Z should preserve information needed to reconstruct X
  - however, preserving too much information may lead to trivial copying rather than meaningful compression

Data Processing Inequality

- for the autoencoder pipeline $X \to Z \to \hat{X}$,
  - the variables from a Markov chain, so the data processing inequality gives

$$I(X; \hat{X}) \le I(X; Z) \le H(X)$$

- interpretation
  - $I(X; Z) \le H(X)$:
    - the latent code cannot contain more information about X than the input itself
  - $I(X; \hat{X}) \le I(X; Z)$:
    - the decoder cannot recover information that is not already present in Z
  - once information is discarded by the encoder, it cannot be recreated downstream

- implications for autoencoders
  - the bottleneck limits how much information can pass from X to Z
  - a smaller bottleneck usually creates stronger compression pressure
  - the challenge is to discard redundancy while preserving structure needed for reconstruction

- goal of representation learning: $Z = f_\theta(X)$
  - should preserve as much useful information as possible under limited latent capacity

Dimensionality Reduction and Information Preservation

- general setup

$$f: \mathbb{R}^d \to \mathbb{R}^k, \qquad Z = f(X)$$

  - in classical dimensionality reduction, typically $k \ll d$
  - more generally, Z is a lower-capacity representation of the input

- what makes a representation Z "good?
  - retain enough information about the input: $I(X;Z)$ should be large
  - remain compact: Z should use limited capacity
  - support downstream tasks: if a task variable Y exists, $I(Z;Y)$ should be large

- trade-off
  - these objectives cannot usually be maximized simultaneously
  - stronger compression may discard useful information
  - different methods realize different trade-offs between compactness and utility

- autoencoder perspective
  - in autoencoders, the main concern is preserving information required for reconstructing X

Autoencoder Architecture

- basic structure

$$X \to Z \to \hat{X}$$

$$Z = f_\theta(X) \in \mathbb{R}^k, \qquad \hat{X} = g_\phi(Z) \in \mathbb{R}^d$$

  - the encoder maps the input to a latent code
  - the bottleneck limits latent capacity
  - only information contained in Z can be used for reconstruction

- reconstruction loss
  - mean squared error $\mathcal{L}_{rec} = \mathbb{E}_{p_{data}(x)}\left[\left|\left|X - \hat{X}\right|\right|^2\right]$
  - binary cross-entropy (Bernoulli assumption):

$$\mathcal{L}_{rec} = -\mathbb{E}\left[X \log \hat{X} + (1 - X)\log(1 - \hat{X})\right]$$

  - from an information-theoretic view, the reconstruction loss acts as a distortion measure

Autoencoder Architecture

- probabilistic interpretation
  - if the decoder defines a conditional distribution $p_\phi(x|z)$, then reconstruction training can be viewed as maximizing the conditional likelihood

$$\mathcal{L}_{rec} \propto -\mathbb{E}_{p(x)}\left[\log p_\phi(x|z)\right]$$

- information-theoretic interpretation
  - minimizing reconstruction loss encourages the latent code Z to preserve information useful for reconstructing X
  - under a bottleneck constraint, the model must keep salient structure while discarding less useful variation

Capacity and Regularization

- bottleneck dimension and capacity
  - let $x \in \mathbb{R}^d$ and $z \in \mathbb{R}^k$
  - undercomplete ($k < d$): the latent code has lower dimension, creating explicit compression
  - overcomplete ($k \geq d$): dimensionality alone does not enforce compression, so additional constrains are needed

- why can overcomplete autoencoders fail?
  - with sufficient capacity, the network may converge to an identity mapping $f_\theta \approx g_\phi^{-1}$
  - information-theoretically, $I(X;Z)$ can approach $H(X)$, so little or no meaningful compression occurs

- regularization as a soft bottleneck
  - regularization can restrict effective information flow even when k is not small
  - sparse AE encourages only a few latent units to be active ($\mathcal{L} + \lambda||Z||_1$)
  - denoising AE reconstruct clean X from corrupted input $\hat{X}$, encouraging robustness and invariance
  - contractive AE penalizes highly sensitive encoders and promotes locally stable representations ($\mathcal{L} + \lambda\left||J_f|\right|_F^2$; Jacobian penalty)

Information Bottleneck (IB) Principle

- IB objective (Tishby et al., 2000)
  - for input X, target Y, and representation Z
  - the goal is to find the most compression representation that still preserves information relevant to predicting Y

$$\min_{p(z|x)} I(X;Z) - \beta I(Z;Y)$$

- interpretation of the terms
  - $I(X;Z)$: representation complexity or coding rate → minimize
  - $I(Z;Y)$: task-relevant information → maximize
  - β: trade-off parameter between compression and prediction

- why is there a limit?
  - if Z is derived from X, then $Y \leftrightarrow Z \leftrightarrow X$ and the data processing inequality implies ⇒ $I(Z;Y) \leq I(X;Y)$
  - no representation can contain more information about Y than the original input X

- connection to autoencoders
  - autoencoders aim to preserve information needed to reconstruct X
  - IB aims to preserve only the information needed to predict Y
  - in a self-supervised setting, we may view autoencoding as a related special case with $Y = X$